ARTIFICIAL INTELLIGENCE MINI PROJECT PRESENTATION

AI-Based Smart Waste Segregation & Classification System

Automated Point-of-Disposal Recognition using MobileNetV2 Transfer Learning, Color-Coded Municipal Guidance, and Empirical Telemetry

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Introduction, Problem & Motivation

Real-World Problem

The Global Waste Crisis

- **2.01 Billion Tonnes** of municipal solid waste generated annually.
- Over **80% is dumped unsegregated** into open landfills.
- **Contamination Bottleneck:** When clean recyclables (paper, plastic) mix with wet food scraps, they become non-recyclable.

The Solution: Point-of-Disposal AI

- **Eliminates Manual Sorting:** Protects sanitation workers from bio-aerosols, pathogens, and sharp glass/metal.
- **Instant Citizen Guidance:** Replaces "wish-cycling" with immediate visual classification and bin directions.
- **Scalable Circular Economy:** Maximizes purity of recycling streams right at the disposal container.

Simple Real-World Example: A student holding a used beverage cup or cardboard box is unsure which bin to use. Using our system's live camera, the item is classified in < 50ms, showing the exact bin color and whether to clean it first.

Main Project Objectives

Project Scope

1. Dataset Engineering

Curate 1,647 photographic samples across 5 balanced classes and apply aspect-ratio preserving crop to prevent item squashing.

2. Deep Learning Model

Fine-tune MobileNetV2 with depthwise separable convolutions to achieve ≥ 85% test accuracy and < 50ms inference.

3. Disposal Directives

Generate authoritative disposal instructions matched to the predicted class with designated municipal bin colors.

4. Empirical Dashboard

Build a live Chart.js statistics dashboard tracking real database records without hardcoded numbers.

5. Persistent History

Store record IDs, timestamps, confidence scores, and preview thumbnails in an SQLite relational database.

6. Safety & Awareness

Enforce a 60% confidence warning threshold and deploy a 5-module environmental awareness guide.

Existing Work & Literature Review

Literature Review

Authors & Year	Proposed Approach	Key Advantage	Identified Limitation / Gap
Yang & Thung (TrashNet, 2016)	SVM & Softmax with SIFT/HOG	Pioneered open benchmark dataset	Handcrafted features fail on crushed/deformed waste (~63% acc)
Bircanoğlu et al. (RecycleNet, 2018)	ResNet-50 & Inception-v4	Demonstrated deep CNN accuracy	Massive parameter count (> 25M); too slow for edge bins
Ruiz et al. (2019)	Standard CNNs (VGG, MobileNet)	Evaluated varied lighting conditions	Standard resizing squashes aspect ratio of cylindrical bottles
Sadhana et al. (2022)	MobileNet IoT Segregator	Raspberry Pi edge implementation	Coarse 3-class system (Wet/Dry/Metal); no safety confidence checks

Our Research Contribution: We bridge these gaps with a fine-tuned 5-class MobileNetV2 architecture utilizing aspect-ratio preservation, a 60% low-confidence warning threshold, and real-time SQLite database analytics.

Proposed System Architecture & Workflow

System Architecture

[1. INPUT] Camera Stream (WebRTC) OR Image File Upload (JPG/PNG/WEBP)
↓
[2. PREPROCESSING] RGB Standardization → ImageOps.fit(224, 224) → Normalize [-1, 1]
↓
[3. AI INFERENCE] MobileNetV2 (53 Layers + GAP + Dropout 0.3 + Dense 5 Softmax)
↓
[4. SAFETY LOGIC] Check Confidence Threshold (≥ 60% vs. < 60%)
- If ≥ 60%: High Confidence Result + Standardized Disposal Directive + Bin Color
- If < 60%: Amber Warning Alert ("AI not confident") + "Try Another Image"

↓
[5. PERSISTENCE] SQLite Database Archiving (Record ID, Class, Confidence, Thumbnail, Time)
↓
[6. TELEMETRY] Real-Time Waste Statistics Dashboard (Interactive Chart.js Visualizations)

Dataset & Preprocessing Pipeline

Data Engineering

Curated Dataset: 1,647 Images

- **Plastic (324)** Water bottles, detergent jugs, food containers
- **Paper (318)** Cardboard boxes, newspapers, packaging cartons
- **Metal (342)** Soda cans, tin food cans, foil packaging
- **Glass (328)** Clear & amber bottles, culinary jars, vials
- **Organic (335)** Fruit peels, leftover food, tea grounds

Preprocessing Innovations

- **Aspect-Ratio Preserving Crop:** Standard resizing squashes elongated bottles into squares. We use `ImageOps.fit` with bicubic interpolation to preserve natural silhouettes.
- **Mathematical Normalization:** Scaled raw pixels [0, 255] to [-1.0, 1.0].
- **Data Augmentation:** Random rotations (±20°), zoom, shifts, and horizontal flips during training to combat overfitting.

Data Split: 80% Training ($1,318$ images) • 20% Isolated Unseen Testing (329 images)

Methodology & AI Deep Learning Technique

AI Model Architecture

Why MobileNetV2?

- **Depthwise Separable Convolutions:** Splits standard convolution into depthwise and pointwise operations, cutting computation by **~85%**.
- **Inverted Residual Bottlenecks:** Preserves complex gradient flow while maintaining lightweight memory footprint.
- **Compact File Size:** Just **27.61 MB** with 2.2 million parameters.
- **Sub-50ms Latency:** Executes real-time inference on commodity CPUs and Raspberry Pi.

Custom Head & Two-Phase Training

- **Global Average Pooling:** Reduces 7x7x1280 feature maps to a 1280 vector.
- **Dense + Dropout(0.3):** Dense projection with regularization to prevent overfitting.
- **Softmax Output:** Computes calibrated probabilities across all 5 waste classes.
- **Phase 1 (Transfer):** 10 epochs with frozen base layers ($\eta = 10^{-3}$).
- **Phase 2 (Fine-Tuning):** 15 epochs unfreezing top 30 layers with cosine decay ($\eta = 10^{-4}$).

Results & Performance Evaluation

Experimental Results

88.02%
UNSEEN TEST ACCURACY

0.2647
TEST CROSS-ENTROPY LOSS

42 ms
MEAN INFERENCE LATENCY

Class	Precision	Recall	F1-Score	Key Performance Insight
Organic	0.96	0.97	**0.96**	Highest recall due to distinct biological textures (peels, seeds)
Metal	0.92	0.89	**0.90**	High precision driven by specular reflective surface highlights
Paper	0.89	0.91	**0.90**	Strong detection of matte cellulose textures and cardboard folds
Glass	0.87	0.86	**0.86**	Accurate jar/bottle contours; minor reflection overlap with plastic
Plastic	0.86	0.84	**0.85**	Robust identification across varied transparent & opaque bottles

System Features & Live Demonstration

Implemented System

1. Standardized Disposal Directives

Displays exact instructions matched to the predicted class (e.g., "*Clean the item if possible and place it in the plastic recycling bin*") with designated municipal bin badges.

2. Waste Statistics Dashboard

Live Chart.js bar and doughnut charts computing stream counts and percentage distributions strictly from actual SQLite database records (no fake data).

3. Classification History

Archives records newest-first with thumbnail previews, category pills, confidence scores, timestamps, and a modal-confirmed clear history function.

4. Low-Confidence Safety Warning

When confidence < 60%, displays an amber warning alert with a "Try Another Image" button, keeping predictions transparent.

Conclusion & Future Scope

Future Enhancements

Project Summary & Accomplishments

- Successfully engineered a real-time, lightweight AI waste classification system achieving **88.02% test accuracy**.
- Delivered an end-to-end web platform bridging computer vision classification with actionable recycling instructions.
- Eliminates sorting guesswork at the point of disposal to preserve clean recycling streams.

Practical Future Extensions

- **YOLOv8 Multi-Object Detection:** Localize and segment multiple overlapping waste items in a single photo.
- **IoT Hardware Actuation:** Connect Flask API with an ESP32 or Raspberry Pi to physically open matching bin lids via servo motors.
- **Offline Mobile App:** Export to TensorFlow Lite (TFLite) for standalone deployment on Android/iOS devices.
- **Ultrasonic Fill Monitoring:** Alert sanitation staff when bin compartments reach 90% capacity.

References & Live System Demonstration

Thank You

Key Academic References

- **Howard et al. (2018):** *MobileNetV2: Inverted Residuals and Linear Bottlenecks*, IEEE CVPR.
- **Yang & Thung (2016):** *Classification of Trash for Recyclability Status*, Stanford CS229.
- **Chu et al. (2020):** *Multilayer Hybrid Deep-Learning System for Household Waste Classification*, IEEE TII.
- **World Bank (2018):** *What a Waste 2.0: Global Solid Waste Management to 2050*.

Live System Demonstration Links

- **Home Page:** `http://127.0.0.1:5000/`
- **Live Classifier:** `http://127.0.0.1:5000/classify`
- **Statistics Dashboard:** `http://127.0.0.1:5000/dashboard`
- **Classification History:** `http://127.0.0.1:5000/history`
- **Environmental Awareness:** `http://127.0.0.1:5000/awareness`

Thank You! Any Questions?

We welcome questions from the evaluation committee.